

in the area of Apollo C&W design indicates that the ideal C&W systems for future spacecraft would have the following capabilities: an acknowledge mode, dedicated resets/inhibits for each C&W channel, a memory or latching system for identifying the source of short-term abnormalities, a variable time delay for screening transients in individual C&W channels, and the capability to alter the alarm limits easily. For the last two features, trade-off studies should be performed to determine the advantages between onboard and ground adjustment schemes.

A special-purpose bank of annunciator lights was used on the CM to display the status of the launch vehicle during boost. Eight lights, grouped within a 5.72- by 8.26-centimeter (2.25 by 3.25 inch) area adjacent to the commander's flight instruments, provided the fundamental booster status information required by the crewmembers during launch. By means of this display, the crewmembers could determine whether or not each booster engine was developing enough thrust, whether or not the booster guidance was functioning properly, and whether or not the staging sequence was proper.

A unique annunciator was the LM lunar-contact light, which illuminated when 3-meter (10 foot) long probes on the LM landing gear contacted the lunar surface to alert the flight crew to shut down the descent engine. These two lights, installed on each side of the panel, were also provided to compensate for a possible loss of external vision because of the formation of a dustcloud caused by the landing engine thrust. Each light was round and approximately 2.5 centimeters (1 inch) in diameter. For distinction from all other annunciators, each light was equipped with a blue lens.

Special flight instruments. - Several unique displays were developed especially for Apollo spacecraft. Among these were the CM and LM flight director attitude indicator (FDAI) devices, the CM EMS, the LM range/range-rate meter, and the DSKY combinations for the primary guidance, navigation, and control system (PGNCS) computers.

Flight director attitude indicator: The FDAI, the primary flight display in both the LM and the CM, integrated into a single instrument the display of vehicle attitude, rotation rate, and attitude error. The display design was similar to that used for aircraft attitude and flight director indicators, except that attitude, errors, and rates were displayed for all three vehicle axes. Attitude was displayed on a sphere marked in an "inside looking out" fashion. Errors and rates were displayed with "fly to" needles. Attitude could be displayed to 1° , and the scaling for the error and rate needles could be changed to suit the flight situation.

Numerous changes were made to the attitude indicators before and during production and after the initial Apollo flights. Changes were made for a variety of reasons. To obtain more precise monitoring of vehicle attitude during active maneuvers, 5° yaw markings and 1° pitch markings were added. This requirement for improved attitude resolution resulted from Gemini reentry-targeting experience. At approximately the same time, as a result of recently acquired experience in both CM and LM flight simulations, astronauts expressed the desire to eliminate unnecessary dissimilarities in attitude sphere markings. The indicators in both vehicles were changed accordingly. Later, changes were made to the LM FDAI error and rate needle scaling